

Executive Business Report

Intelligent Sales Forecasting and Demand Analysis for Retail Inventory Management

Executive Summary

This report presents the results of a sales forecasting and demand analysis system developed using four years of Superstore sales data. The objective was to help improve inventory planning, reduce stock shortages, and avoid excess inventory. Multiple forecasting models were evaluated, with XGBoost delivering the best prediction accuracy. The analysis also identified unusual sales patterns, grouped products based on demand behavior, and highlighted areas where inventory planning can be improved. These findings provide practical insights that can support better stocking decisions and overall supply chain efficiency.

Key Findings from Data Exploration and Forecasting

The sales analysis revealed that different product categories and regions contribute differently to overall revenue. Office Supplies consistently generated steady sales, while Technology products showed high sales during seasonal periods. Monthly sales displayed a clear upward trend over the four-year period, with noticeable increases during November and December, indicating strong holiday season demand.

Three forecasting models—SARIMA, Prophet, and XGBoost—were compared. Their performance is summarized below:

Model	MAE	RMSE	MAPE
SARIMA	19,244.49	19,950.07	20.53%
Prophet	20,250.79	22,318.41	21.86%
XGBoost	14,763.81	18,337.41	14.48%

XGBoost achieved the lowest prediction error across all evaluation metrics and was selected as the final production model.

Three-Month Sales Forecast

Based on the XGBoost forecasting model, the expected sales for the next three months are:

Forecast Month	Predicted Sales
Month 1	86,530
Month 2	87,955
Month 3	88,256

The forecasts suggest that sales will remain relatively stable over the next three months, with only moderate month-to-month variation. Since the model achieved a Mean Absolute Percentage Error (MAPE) of approximately **14.5%**, actual sales are expected to remain reasonably close to these predicted values under normal business conditions.

Top Three Sales Anomalies

The anomaly detection process identified several weeks where sales were unusually different from normal patterns.

- 1. March 22, 2015**
 - Sales increased sharply to approximately **37,704**.
 - Likely caused by promotional campaigns, bulk customer purchases, or seasonal demand.
- 2. December 2, 2018**
 - Weekly sales reached approximately **35,999**.
 - Most likely associated with year-end holiday shopping and discount events.
- 3. November 18, 2018**
 - Sales increased to approximately **30,572**.
 - Likely influenced by festive sales events such as Black Friday or other promotional campaigns.

These unusual sales periods should be considered when planning future inventory levels.

Product Demand Segmentation

Using K-Means clustering, products were grouped according to their sales volume, growth rate, volatility, and average order value.

Cluster 1 – High Volume Products

Examples include:

- Chairs
- Phones
- Storage
- Binders

Recommended Strategy:

Maintain high inventory levels with continuous replenishment to avoid stock shortages.

Cluster 2 – Stable Demand Products

Examples include:

- Appliances
- Paper
- Art
- Furnishings

Recommended Strategy:

Maintain moderate inventory levels using regular replenishment cycles.

Cluster 3 – Premium Products

Examples include:

- Copiers
- Machines

Recommended Strategy:

Maintain limited stock due to high value while ensuring availability for important customer orders.

Cluster 4 – Fast Growing Products

Example:

- Supplies

Recommended Strategy:

Increase inventory gradually while closely monitoring demand growth to prevent shortages.

Business Recommendations

1. Increase inventory before seasonal demand peaks.

Sales consistently rise during November and December. Stocking additional inventory before these months can reduce lost sales and improve customer satisfaction.

2. Use XGBoost forecasts for monthly inventory planning.

The XGBoost model produced the most accurate predictions (MAPE = 14.48%), making it suitable for supporting purchasing and inventory decisions.

3. Apply different stocking strategies for different product segments.

High-volume products should receive frequent replenishment, while premium and low-demand products should be stocked more conservatively to reduce storage costs and inventory risk.

Risk and Limitation

The forecasting system is based entirely on historical sales data. It cannot predict unexpected market events such as economic changes, competitor actions, supply chain disruptions, or sudden shifts in customer preferences. Therefore, forecast results should be used alongside business knowledge and market insights rather than as the sole basis for decision-making.

Conclusion

The developed sales forecasting system successfully combines forecasting, anomaly detection, and product demand segmentation into a practical decision-support solution. Among the evaluated models, XGBoost demonstrated the highest forecasting accuracy and is recommended for production use. By applying these insights, retail businesses can improve inventory planning, reduce operational costs, minimize stock shortages, and make more informed supply chain decisions.